

Technical Specification Document — Workflow

Integrated Accounting System — Tawaredat Trading Company

An architectural/operational specification describing the project structure, workflow, screen-navigation mechanisms, libraries used, and the data flow between frontend and backend, with security and performance considerations.

1. Overview of Structure and Development Environment

1.1 System Architecture

The system is a **Server-Rendered** web application built on the Django framework (MTV pattern: Model–Template–View). There is no separate SPA frontend; every screen is rendered on the server and delivered as HTML. Consequently, the SPA concepts of "Props / State" map to the following in the Django context:

SPA concept	Equivalent in this system
Props	Template Context variables passed via render()
State (client)	Minimal JS state (menu collapse, partial fetch)
State (server)	Database (Models), Session, Cache
Navigation Source of Truth	The URL (path + query string)

1.2 Project Structure (package layout)

accounting_system/ # project settings |— settings.py # settings (DB, ALLOWED_HOSTS, CELERY, network binding) |— urls.py
root router (includes sub-apps) |— celery.py # Celery setup |— middleware.py # LoginThrottleMiddleware (login attempt throttling)
accounts/ sales/ purchases/ treasury/ assets/ hr/ reports/ documents/ warehouses/ company/ ai_analysis/
concrete_production/ contractors/ backups/ sync/ users/ audit/ budget/ currency/ bank-reconciliation/ notifications/ common/
recurring/ credit_notes/ cheques/ sales_returns/ purchase_returns/ stock_adjustments/ payment_receipts/ # Django apps (each: models, views, urls, templates)
common/ # permissions, context processors, shared models templates/base/ # base template (sidebar / topbar)
static/ staticfiles/ media/ requirements.txt network.conf run_multi.py run.bat

Each "app" contains: models.py, views.py, urls.py, and a templates// folder.

1.3 Development & Runtime Environment

- **Language:** Python 3.15 (interpreter: D:\python 3.15\python.exe)
- **Framework:** Django 4.2.30 (LTS)
- **Database:** SQLite by default; switchable to PostgreSQL via
DJANGO_DB_ENGINE=postgres + connection settings.
- **UI:** Bootstrap 5.3 + Cairo Font + Font Awesome 6.4 + Jazzmin (admin theme).
- **Runtime:** run_multi.py (multi-address binding) for development, or
Nginx + Gunicorn in production (see deploy/).

- **Async tasks:** Celery 5.6.3 (eager mode by default, no broker needed).
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2. Workflow

2.1 General Flow — from Login to Completing the Core Operation

1. **Authentication:** the user reaches `/accounts/login/`; Django authenticates, creates a session, and redirects (`?next=`) to the dashboard `/`.
2. **Dashboard:** shows indicators (via `dashboard_view` + reports); navigation via the permission-aware sidebar.
3. **Core operation — Concrete Sales Invoice cycle:**
 - a. `concrete_production:customer_request_create` ← customer request.
 - b. `concrete_production:production_order_create` ← production order linked to the request.
 - c. recording batches (`production_batch`) and deliveries (`delivery_schedule`).
 - d. on production order completion (`status='completed'`) a **Sales Invoice** is auto-created and linked (`SalesInvoice.production_order`) inside `transaction.atomic`, and its journal entry is posted.
 - e. `sales:sales_invoice_post` ← posts the journal entry (updates account balances atomically via `select_for_update`).
 - f. `sales:sales_invoice_print` / `whatsapp` ← print / send.
4. **Reports:** aggregated in reports (cached via `@cache_page`).

2.2 Inter-step Linking Mechanism

Each step ends with a `redirect()` to the next screen using `reverse()` with the route name and the primary key (`pk`) in the path — keeping the **URL** as the **Single Source of Truth** for navigation state:

```
# concrete_production/views.py — after scheduling return redirect('concrete_production:production_order_detail', pk=order.pk)
```

3. Screen Navigation Mechanisms

3.1 Internal Routing

- The root router `accounting_system/urls.py` includes apps via `include()` with an `app_name` (namespace) to avoid name collisions.
- Each app has a `urls.py` defining patterns and path converters:

```
# sales/urls.py app_name = 'sales' urlpatterns = [ path('invoices/', views.sales_invoice_list, name='invoice_list'), path('invoices/create/', views.sales_invoice_create, name='invoice_create'), path('invoices/<uuid:pk>/', views.sales_invoice_detail, name='invoice_detail'), path('invoices/<uuid:pk>/post/', views.sales_invoice_post, name='invoice_post'), ]
```

extracts the primary key from the URL and passes it to the view.

3.2 Template Navigation

Templates use the `{% url %}` tag with the route name and namespace (never hard-coded URLs):

```
<a href="{% url 'sales:invoice_create' %}">Sales Invoice</a>
```

Highlighting the active screen by comparing `request.resolver_match.url_name`:

```
<a href="{% url 'dashboard' %}" class="{% if request.resolver_match.url_name == 'dashboard' %}active{% endif %}"> Dashboard</a>
```

Passing data between screens via **query string** — example from the sidebar:

```
<a href="{% url 'payment_receipts:create' %}?type=receipt">Receipt Voucher</a>
```

Read in the view via `request.GET.get('type')`.

3.3 "Props / State" in the Django Context

- **Props** = template context:

```
return render(request, 'concrete_production/production_order_detail.html', { 'order': order, 'batches': batches, 'deliveries': deliveries, 'costs': costs, })
```

- **State** — Database (entities linked across screens): relationships (FK) link

screens. Example linking an invoice to a production order:

```
# sales/models.py class SalesInvoice(models.Model): production_order = models.ForeignKey('concrete_production.ProductionOrder', null=True, blank=True, on_delete=models.SET_NULL, related_name='sales_invoice')
```

Shown in the order-detail template as: `{{ order.sales_invoice.invoice_number }}`.

- **State** — Session: branch / user data.
- **State** — Context Processors: injected into every template automatically:

```
# common/context_processors.py def user_permissions_context(request): if request.user.is_superuser: perms = {'all'} else: perms = set(request.user.get_all_permissions()) return {'user_perms': perms, 'is_superuser': request.user.is_superuser}
```

Used to show/hide sidebar links by permission:

```
{% if "all" in user_perms or "sales.add_salesinvoice" in user_perms %} <a href="{% url 'sales:invoice_create' %}">Sales Invoice</a>{% endif %}
```

- **State** — Messages Framework: one-time notifications between requests

(`messages.success(request, '...')`).

- **State** — Client (JS): minimal. Example fetching partial data safely:

```
const data = JSON.parse(document.getElementById('chart-data').textContent);
```

injected via `{{ chart_data_json|json_script:"chart-data" }}` (avoids XSS vs `|safe`).

4. Software and Libraries (versions and rationale)

Library	Version	Role / reason
Python	3.15	Runtime interpreter
Django	4.2.30 (LTS)	Web framework; long-term support, PostgreSQL ready

Library	Version	Role / reason
psycopg2-binary	>=2.9	PostgreSQL driver when switching from SQLite
django-crispy-forms	2.5	Elegant form rendering with Bootstrap
crispy-bootstrap5	2026.3	Bootstrap 5 support for crispy-forms
django-jazzmin	>=2.6	Improved Django admin UI (RTL aware)
reportlab	>=4.0	PDF reports / invoices
openpyxl	>=3.1	Excel import / export
python-dateutil	>=2.8	Date handling in reports
faker	>=20.0	Dummy data for tests (seed_dummy_data)
Celery	5.6.3	Async tasks (balance recalculation)
redis (py)	8.0.1	Celery broker / result store (optional)
Bootstrap 5.3 / Font Awesome 6.4 / Cairo font	—	Arabic RTL interface

5. Data Flow (Frontend ↔ Backend)

5.1 Textual and Visual Data-Flow Diagram

A visual diagram (image + PDF) is available at: docs/diagrams/data_flow.svg and docs/diagrams/data_flow.pdf — illustrating the same flow below visually.

[Browser] | GET /sales/invoices/<pk>/ (or POST form + csrfmiddlewaretoken) ▼ [Server: Nginx ← Unicorn / runserver] | (prod: Nginx listens on addresses, proxies to 127.0.0.1:8012) ▼ [Middleware] → LoginThrottleMiddleware, SessionMiddleware, CsrfViewMiddleware ▼ [URL Resolver] → matches path, calls View with params (pk) ▼ [View] → permission check (permission_required_or_login) | → ORM query (select_related / prefetch_related) | ▼ ▼ [Database: SQLite / PostgreSQL] [Context] ← builds context dict (objects + context processors) ▼ [Template] ← renders HTML (DTL) + static assets ▼ [HTTP Response] → browser renders the screen

For partial data (JSON APIs) the View returns JsonResponse consumed by the browser via fetch:

```
# concrete_production/views.py
def api_silo_stock(request):
    silos = Silo.objects.filter(is_active=True)
    data = [{'id': s.id, 'code': s.code, 'stock': float(s.current_stock_tons)} for s in silos]
    return JsonResponse({'silos': data})
```

```
fetch('/concrete/api/silo-stock/').then(r => r.json()).then(d => renderChart(d.silos));
```

5.2 Security Considerations

- **CSRF:** CsrfViewMiddleware + {% csrf_token %} in every POST form; and CSRF_TRUSTED_ORIGINS derived from the bind addresses.
- **Permissions:** @permission_required_or_login + object-level check (has_object_permission) to block access to other branches.
- **ALLOWED_HOSTS:** derived from bind addresses (SERVER_BIND_HOSTS).
- **SECRET_KEY:** mandatory in production (raises if missing).
- **Login throttling:** LoginThrottleMiddleware (10 attempts / 5 min per IP → 429).
- **JSON injection:** json_script instead of |safe.

- **Secure cookies** (Secure/HSTS) when SSL is enabled via `DJANGO_SECURE_SSL_REDIRECT`.

5.3 Performance Considerations

- **Avoid N+1:** `select_related/prefetch_related` — example:

`ProductionOrder.objects.select_related('customer_request__customer','mix_design').`

- **Pagination:** `Paginator(queryset, 25)` with `?page=` switching.
- **Caching:** `@cache_page(300) + @vary_on_cookie` for heavy screens.
- **Heavy tasks:** balance recalculation via a Celery task (`recalc_balances_task`)

instead of synchronous execution.

- **Concurrency safety:** critical stock updates inside `transaction.atomic()` + `select_for_update()` (e.g., `SiloTransaction.save`, `JournalEntry.post`) to avoid lost updates.

6. Screen Permission Matrix

The table below was extracted automatically from the view decorators across all apps (covers 337 screens/functions). The symbol — (public/no restriction) means no permission decorator was found (some are protected by `login_required` only, and some API endpoints are protected by an API key instead of a Django permission). The full matrix is also saved in `docs/permissions_matrix.md`.

App	View (function)	Required access / permission
accounts	account_list	Permission: 'accounts.view_account'
accounts	account_create	Login required
accounts	account_detail	Login required
accounts	account_edit	Login required
accounts	account_statement	Login required
accounts	journal_list	Permission: 'accounts.view_journalentry'
accounts	journal_create	Permission: 'accounts.add_journalentry'
accounts	journal_detail	Login required
accounts	journal_post	Login required
accounts	trial_balance	Login required
accounts	chart_of_accounts	Login required
accounts	export_accounts	Login required
accounts	import_accounts	Login required
accounts	export_journal	Login required
accounts	fiscal_year_close	Login required
ai_analysis	dashboard	Login required
ai_analysis	analyze_error	Login required

App	View (function)	Required access / permission
ai_analysis	auto_detect	Login required
ai_analysis	error_history	Login required
ai_analysis	error_detail	Login required
ai_analysis	apply_solution	HTTP method restricted: require_POST
ai_analysis	api_detect_errors	HTTP method restricted: require_POST
ai_analysis	api_analyze_error	HTTP method restricted: require_POST
ai_analysis	api_error_stats	Login required
assets	asset_list	Login required
assets	asset_create	Login required
assets	asset_detail	Login required
assets	asset_edit	Login required
assets	depreciation_create	Login required
assets	asset_category_list	Login required
assets	asset_category_create	Login required
assets	export_assets	Login required
assets	import_assets	Login required
audit	audit_log_list	Login required
backups	_ensure_backup_dir	— (دقيق الـب/م/ع)
backups	_format_size	— (دقيق الـب/م/ع)
backups	_safe_extract_zip	— (دقيق الـب/م/ع)
backups	backup_dashboard	Login required
backups	create_backup	HTTP method restricted: require_POST
backups	download_backup	Login required
backups	delete_backup	HTTP method restricted: require_POST
backups	restore_backup	Login required
backups	export_json	Login required
backups	import_json	Login required
backups	backup_settings_view	Login required
bank_reconciliation	reconciliation_dashboard	Login required
bank_reconciliation	session_create	Login required
bank_reconciliation	session_detail	Login required
bank_reconciliation	item_list	Login required
bank_reconciliation	item_create	Login required
bank_reconciliation	item_match	Login required
bank_reconciliation	import_csv	Login required
bank_reconciliation	auto_match	Login required
bank_reconciliation	item_delete	HTTP method restricted: require_POST
budget	cost_center_list	Login required

App	View (function)	Required access / permission
budget	cost_center_create	Login required
budget	budget_list	Login required
budget	budget_create	Login required
cheques	cheque_list	Login required
cheques	cheque_create	Login required
cheques	cheque_detail	Login required
cheques	cheque_update_status	HTTP method restricted: require_POST
cheques	cheque_delete	HTTP method restricted: require_POST
cheques	cheque_dashboard	Login required
common	whatsapp_webhook	HTTP method restricted: require_http_methods
company	company_settings	Login required
company	branch_create	Login required
company	branch_edit	Login required
company	branch_delete	HTTP method restricted: require_POST
company	admin_settings_dashboard	Permission: 'accounts.change_accounttype', raise_exception=True
company	account_type_create	HTTP method restricted: require_http_methods
company	account_type_update	HTTP method restricted: require_http_methods
company	account_type_delete	HTTP method restricted: require_http_methods
company	product_create	HTTP method restricted: require_http_methods
company	product_update	HTTP method restricted: require_http_methods
company	product_delete	HTTP method restricted: require_http_methods
company	category_create	HTTP method restricted: require_http_methods
company	category_update	HTTP method restricted: require_http_methods
company	category_delete	HTTP method restricted: require_http_methods
company	unit_create	HTTP method restricted: require_http_methods
company	unit_update	HTTP method restricted: require_http_methods
company	unit_delete	HTTP method restricted: require_http_methods
credit_notes	credit_note_list	Login required
credit_notes	credit_note_create	Login required
credit_notes	credit_note_detail	Login required
credit_notes	credit_note_post	HTTP method restricted: require_POST
credit_notes	credit_note_delete	HTTP method restricted: require_POST
currency	currency_list	Login required
currency	currency_create	Login required
currency	currency_edit	Login required
currency	exchange_rate_history	Login required
documents	document_type_list	Login required
documents	document_type_create	Login required

App	View (function)	Required access / permission
documents	document_type_edit	Login required
documents	document_template_list	Login required
documents	document_template_create	Login required
documents	document_template_edit	Login required
documents	document_list	Login required
documents	document_create	Login required
documents	document_detail	Login required
documents	document_edit	Login required
documents	document_action	Login required
documents	document_add_attachment	Login required
hr	employee_list	Permission: 'hr.view_employee'
hr	employee_create	Permission: 'hr.add_employee'
hr	employee_detail	Permission: 'hr.view_employee'
hr	employee_edit	Permission: 'hr.change_employee'
hr	attendance_list	Permission: 'hr.view_attendance'
hr	attendance_create	Permission: 'hr.add_attendance'
hr	salary_list	Permission: 'hr.view_salary'
hr	salary_create	Permission: 'hr.add_salary'
hr	salary_post	HTTP method restricted: require_POST
hr	department_list	Permission: 'hr.view_department'
hr	department_create	Permission: 'hr.add_department'
hr	export_employees	Permission: 'hr.view_employee'
hr	export_salaries	Permission: 'hr.export_salary'
hr	import_employees	Permission: 'hr.add_employee'
notifications	notification_dashboard	Login required
notifications	template_list	Login required
notifications	template_create	Login required
notifications	send_test_notification	Login required
payment_receipts	receipt_list	Permission: 'treasury.view_bank'
payment_receipts	receipt_create	Login required
payment_receipts	receipt_detail	Login required
payment_receipts	receipt_post	HTTP method restricted: require_POST
payment_receipts	receipt_delete	HTTP method restricted: require_POST
payment_receipts	receipt_print	Login required
payment_receipts	get_customer_invoices	Login required
payment_receipts	get_supplier_invoices	Login required
purchases	supplier_list	Permission: 'purchases.view_supplier'
purchases	supplier_create	Permission: 'purchases.add_supplier'

App	View (function)	Required access / permission
purchases	supplier_detail	Permission: 'purchases.view_supplier'
purchases	supplier_edit	Permission: 'purchases.change_supplier'
purchases	product_list	Login required
purchases	product_create	Login required
purchases	product_edit	Login required
purchases	catalog_settings	Login required
purchases	purchase_invoice_list	Login required
purchases	purchase_invoice_create	Login required
purchases	purchase_invoice_detail	Login required
purchases	purchase_invoice_post	HTTP method restricted: require_POST
purchases	purchase_invoice_print	Login required
purchases	purchase_invoice_whatsapp	HTTP method restricted: require_POST
purchases	supplier_statement_whatsapp	HTTP method restricted: require_POST
purchases	export_suppliers	Login required
purchases	import_suppliers	Login required
purchases	export_products	Login required
purchases	import_products	Login required
purchases	product_barcode_print	Login required
purchases	product_barcode_batch	Login required
purchases	product_price_list	Login required
purchase_returns	purchase_return_list	Permission: 'purchases.view_purchaseinvoice'
purchase_returns	purchase_return_create	Login required
purchase_returns	purchase_return_detail	Login required
purchase_returns	purchase_return_post	HTTP method restricted: require_POST
purchase_returns	purchase_return_delete	HTTP method restricted: require_POST
recurring	recurring_list	Login required
recurring	recurring_create	Login required
recurring	recurring_edit	Login required
recurring	recurring_execute	Login required
recurring	recurring_toggle	Login required
recurring	recurring_delete	Login required
recurring	_save_lines	— (دقيق الـب/م/اع)
reports	_safe_parse_date	— (دقيق الـب/م/اع)
reports	_validate_date_range	— (دقيق الـب/م/اع)
reports	_get_date_range	— (دقيق الـب/م/اع)
reports	dashboard_view	Login required
reports	financial_dashboard	Permission: 'reports.view_reporttemplate'
reports	_age_bucket	— (دقيق الـب/م/اع)

App	View (function)	Required access / permission
reports	workflow_tracker	Permission: 'reports.view_reporttemplate'
reports	report_list	Login required
reports	income_statement	Login required
reports	balance_sheet	Login required
reports	trial_balance_report	Login required
reports	vat_return	Login required
reports	withholding_tax_report	Login required
reports	supplier_report	Login required
reports	supplier_detail_report	Login required
reports	customer_report	Login required
reports	customer_detail_report	Login required
reports	profit_margin_report	Login required
reports	asset_schedule	Login required
reports	payroll_report	Login required
reports	export_report	Login required
reports	_export_simple_xlsx	— (دقيق الب/م/ع)
reports	__init__	— (دقيق الب/م/ع)
reports	__getattr__	— (دقيق الب/م/ع)
reports	_export_daily_sales	— (دقيق الب/م/ع)
reports	_export_daily_purchases	— (دقيق الب/م/ع)
reports	_export_ar_aging	— (دقيق الب/م/ع)
reports	_export_ap_aging	— (دقيق الب/م/ع)
reports	_export_income_statement	— (دقيق الب/م/ع)
reports	_export_balance_sheet	— (دقيق الب/م/ع)
reports	_export_vat_return	— (دقيق الب/م/ع)
reports	_export_payroll	— (دقيق الب/م/ع)
reports	_export_inventory	— (دقيق الب/م/ع)
reports	_export_profit_margin	— (دقيق الب/م/ع)
reports	_export_withholding_tax	— (دقيق الب/م/ع)
reports	_export_supplier_report	— (دقيق الب/م/ع)
reports	_export_customer_report	— (دقيق الب/م/ع)
reports	_export_customer_statement	— (دقيق الب/م/ع)
reports	_export_supplier_statement	— (دقيق الب/م/ع)
reports	_export_trial_balance	— (دقيق الب/م/ع)
reports	_export_asset_schedule	— (دقيق الب/م/ع)
reports	_export_cash_flow	— (دقيق الب/م/ع)
reports	ar_aging_report	Login required
reports	ap_aging_report	Login required

App	View (function)	Required access / permission
reports	inventory_report	Login required
reports	customer_statement	Login required
reports	supplier_statement	Login required
reports	daily_sales_report	Login required
reports	daily_purchases_report	Login required
reports	cash_flow_report	Login required
sales	customer_list	Permission: 'sales.view_customer'
sales	customer_create	Permission: 'sales.add_customer'
sales	customer_detail	Permission: 'sales.view_customer'
sales	customer_edit	Permission: 'sales.change_customer'
sales	sales_invoice_list	Permission: 'sales.view_salesinvoice'
sales	sales_invoice_create	Permission: 'sales.add_salesinvoice'
sales	sales_invoice_detail	Login required
sales	sales_invoice_post	HTTP method restricted: require_POST
sales	sales_invoice_print	Login required
sales	sales_invoice_whatsapp	HTTP method restricted: require_POST
sales	customer_statement_whatsapp	HTTP method restricted: require_POST
sales	export_customers	Login required
sales	import_customers	Login required
sales_returns	sales_return_list	Permission: 'sales.view_salesinvoice'
sales_returns	sales_return_create	Login required
sales_returns	sales_return_detail	Login required
sales_returns	sales_return_post	HTTP method restricted: require_POST
sales_returns	sales_return_delete	HTTP method restricted: require_POST
stock_adjustments	adjustment_list	Login required
stock_adjustments	adjustment_create	Login required
stock_adjustments	adjustment_detail	Login required
stock_adjustments	adjustment_approve	HTTP method restricted: require_POST
stock_adjustments	adjustment_delete	HTTP method restricted: require_POST
sync	_get_or_create_machine	— (دقيق الـب/م/اع)
sync	_verify_api_key	— (دقيق الـب/م/اع)
sync	sync_dashboard	Login required
sync	sync_settings_view	Login required
sync	test_connection	Login required
sync	manual_sync	Login required
sync	sync_log_detail	Login required
sync	api_push	HTTP method restricted: require_http_methods
sync	api_pull	HTTP method restricted: require_http_methods

App	View (function)	Required access / permission
sync	api_status	HTTP method restricted: require_http_methods
sync	api_recalculate	HTTP method restricted: require_http_methods
treasury	bank_list	Permission: 'treasury.view_bank'
treasury	bank_create	Permission: 'treasury.add_bank'
treasury	bank_detail	Permission: 'treasury.view_bank'
treasury	bank_transaction_create	Permission: 'treasury.add_banktransaction'
treasury	safe_list	Permission: 'treasury.view_safe'
treasury	safe_create	Permission: 'treasury.add_safe'
treasury	safe_detail	Permission: 'treasury.view_safe'
treasury	safe_transaction_create	Permission: 'treasury.add_safetransaction'
treasury	export_banks	Permission: 'treasury.view_bank'
treasury	export_safes	Permission: 'treasury.view_safe'
treasury	import_banks	Permission: 'treasury.add_bank'
treasury	import_safes	Permission: 'treasury.add_safe'
users	clean_password2	— (ديق الـب/ماع)
users	save	— (ديق الـب/ماع)
users	user_list	Login required
users	user_create	Login required
users	user_edit	Login required
users	user_delete	HTTP method restricted: require_POST
users	group_list	Login required
users	group_create	Login required
users	group_edit	Login required
users	group_delete	HTTP method restricted: require_POST
users	change_password	Login required
warehouses	warehouse_list	Permission: 'warehouses.view_warehouse'
warehouses	warehouse_create	Permission: 'warehouses.add_warehouse'
warehouses	warehouse_edit	Permission: 'warehouses.change_warehouse'
warehouses	warehouse_detail	Permission: 'warehouses.view_warehouse'
warehouses	warehouse_product_add	Permission: 'warehouses.add_warehouseproduct'
warehouses	movement_list	Permission: 'warehouses.view_stockmovement'
warehouses	movement_create	Permission: 'warehouses.add_stockmovement'
warehouses	movement_detail	Permission: 'warehouses.view_stockmovement'
concrete_production	dashboard	Permission: 'concrete_production.view_concretemixdesign'
concrete_production	mix_design_list	Permission: 'concrete_production.view_concretemixdesign'
concrete_production	mix_design_create	Permission: 'concrete_production.add_concretemixdesign'
concrete_production	mix_design_detail	Permission: 'concrete_production.view_concretemixdesign'
concrete_production	mix_design_edit	Permission: 'concrete_production.change_concretemixdesign'

App	View (function)	Required access / permission
concrete_production	customer_request_list	Permission: 'concrete_production.view_customerrequest'
concrete_production	customer_request_create	Permission: 'concrete_production.add_customerrequest'
concrete_production	customer_request_detail	Permission: 'concrete_production.view_customerrequest'
concrete_production	customer_request_confirm	Permission: 'concrete_production.change_customerrequest'
concrete_production	production_cost_per_m3	Permission: 'concrete_production.view_productionorder'
concrete_production	production_order_list	Permission: 'concrete_production.view_productionorder'
concrete_production	production_order_create	Permission: 'concrete_production.add_productionorder'
concrete_production	production_order_detail	Permission: 'concrete_production.view_productionorder'
concrete_production	production_order_schedule	Permission: 'concrete_production.change_productionorder'
concrete_production	batch_list	Permission: 'concrete_production.view_productionbatch'
concrete_production	production_daily	Permission: 'concrete_production.view_productionorder'
concrete_production	batch_create	Permission: 'concrete_production.add_productionbatch'
concrete_production	batch_detail	Permission: 'concrete_production.view_productionbatch'
concrete_production	batch_update_status	Permission: 'concrete_production.change_productionbatch'
concrete_production	truck_list	Permission: 'concrete_production.view_truck'
concrete_production	truck_create	Permission: 'concrete_production.add_truck'
concrete_production	truck_edit	Permission: 'concrete_production.change_truck'
concrete_production	delivery_list	Permission: 'concrete_production.view_deliveryschedule'
concrete_production	delivery_create	Permission: 'concrete_production.add_deliveryschedule'
concrete_production	cost_list	Permission: 'concrete_production.view_productioncost'
concrete_production	cost_create	Permission: 'concrete_production.add_productioncost'
concrete_production	api_mix_components	Permission: 'concrete_production.view_concretemixdesign'
concrete_production	api_available_trucks	Permission: 'concrete_production.view_truck'
concrete_production	api_production_stats	Permission: 'concrete_production.view_productionorder'
concrete_production	silo_dashboard	Permission: 'concrete_production.view_silo'
concrete_production	cement_daily_inventory	Permission: 'concrete_production.view_silotransaction'
concrete_production	silo_list	Permission: 'concrete_production.view_silo'
concrete_production	silo_detail	Permission: 'concrete_production.view_silo'
concrete_production	silo_create	Permission: 'concrete_production.add_silo'
concrete_production	silo_edit	Permission: 'concrete_production.change_silo'
concrete_production	silo_transaction_create	Permission: 'concrete_production.add_silotransaction'
concrete_production	api_silo_stock	Permission: 'concrete_production.view_silo'
contractors	dashboard	Permission: 'contractors.view_contractor'
contractors	contractor_list	Permission: 'contractors.view_contractor'
contractors	contractor_create	Permission: 'contractors.add_contractor'
contractors	contractor_detail	Permission: 'contractors.view_contractor'
contractors	contractor_edit	Permission: 'contractors.change_contractor'
contractors	contract_list	Permission: 'contractors.view_contract'

App	View (function)	Required access / permission
contractors	contract_create	Permission: 'contractors.add_contract'
contractors	contract_detail	Permission: 'contractors.view_contract'
contractors	contract_edit	Permission: 'contractors.change_contract'
contractors	contract_approve	Permission: 'contractors.change_contract'
contractors	contract_close	Permission: 'contractors.change_contract'
contractors	certificate_list	Permission: 'contractors.view_interimcertificate'
contractors	certificate_create	Permission: 'contractors.add_interimcertificate'
contractors	certificate_detail	Permission: 'contractors.view_interimcertificate'
contractors	certificate_approve	Permission: 'contractors.change_interimcertificate'
contractors	certificate_post	Permission: 'contractors.change_interimcertificate'
contractors	payment_list	Permission: 'contractors.view_contractorpayment'
contractors	payment_create	Permission: 'contractors.add_contractorpayment'
contractors	payment_detail	Permission: 'contractors.view_contractorpayment'
contractors	payment_post	Permission: 'contractors.change_contractorpayment'
contractors	api_contract_items	Permission: 'contractors.view_contract'
contractors	api_contractor_stats	Permission: 'contractors.view_contractor'

7. Expanded Production Section (Deployment: Nginx + Gunicorn + systemd/nssm)

7.1 Overview

For production we recommend: ****Nginx**** (reverse proxy + TLS termination) <- ****Gunicorn**** (multiple WSGI workers) <- ****Django****. Static files are collected with collectstatic, and processes are managed as services (systemd on Linux, or NSSM on Windows).

7.2 Gunicorn Setup (backend application)

Basic command:

```
gunicorn accounting_system.wsgi:application --bind 127.0.0.1:8012 --workers 3 --timeout 120
```

Required environment variables (DJANGO_DEBUG=false, DJANGO_SECRET_KEY, DJANGO_ALLOWED_HOSTS, and bind addresses from network.conf). Ready-made script: deploy/start_backend.bat.

7.3 Nginx Setup (reverse proxy)

deploy/nginx_dual.conf listens on the required addresses and proxies to Gunicorn:

```
upstream accounting_backend { server 127.0.0.1:8012; } server { listen 192.168.1.31:8012; listen 196.218.24.45:8012; # if this address is assigned to a host interface; otherwise use 0.0.0.0 and forward from the router server_name _; client_max_body_size 20M; location /static/ { alias J:/2027/accounting_system/staticfiles/; expires 30d; access_log off; } location /media/ { alias J:/2027/accounting_system/media/; expires 7d; } location / { proxy_pass http://accounting_backend; proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr; proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for; proxy_set_header X-Forwarded-Proto $scheme; } }
```

Reload without downtime: `nginx -s reload`.

7.4 systemd Service (Linux)

`/etc/systemd/system/accounting.service`:

```
[Unit] Description=Accounting Gunicorn After=network.target [Service] User=www-data WorkingDirectory=/opt/accounting_system EnvironmentFile=/opt/accounting_system/.env ExecStart=/opt/venv/bin/gunicorn accounting_system.wsgi:application --bind 127.0.0.1:8012 --workers 3 Restart=on-failure [Install] WantedBy=multi-user.target
```

Enable: `systemctl daemon-reload && systemctl enable --now accounting`; and Nginx:

`systemctl enable --now nginx`.

7.5 Windows Service via NSSM

NSSM turns any script into a Windows service:

```
nssm install Accounting "D:\python 3.15\python.exe" "J:\2027\accounting_system\deploy\start_backend.bat" nssm set Accounting AppDirectory "J:\2027\accounting_system" nssm start Accounting
```

To keep Nginx as a service use `nginx-service` or NSSM the same way.

7.6 Static Files and SSL

- Collect static: `python manage.py collectstatic --noinput`.
- SSL: use Certbot (Let's Encrypt) with Nginx; enable `DJANGO_SECURE_SSL_REDIRECT=true` and restart the worker.

7.7 Management and Switch Commands

- Reload Nginx: `nginx -s reload`
- Restart worker: `systemctl restart accounting` / `nssm restart Accounting`
- Change addresses/port: edit `network.conf` then reload/restart the services (no code change needed).